Decision threshold selection and evaluation metrics

"Decision threshold selection for screening or triage prediction models in health, and evaluation metrics for imbalanced outcomes (sensitivity, precision, F1, precision-recall AUC) instead of accuracy. Return a matrix with: authors, year, journal, DOI, clinical context, threshold criterion, metrics recommended, and main conclusion. Limit to 2015-2026." *Generated by LeapSpace*

Decision-threshold selection and imbalance-aware metrics in health screening/triage

Authors	Year	Journal	DOI	Clinical context	Threshold criterion	Metrics recommended	Main conclusion
Not provided in retrieved evidence	2019	*Critical Care*	10.1186/s13054-019-2351-7	ED triage for critical care and hospitalization prediction	Clinical thresholds were explored through decision-curve analysis rather than a fixed default cutoff; operating points were evaluated across the threshold range [1]	AUC plus decision-curve net benefit; the paper emphasizes clinical-threshold evaluation over accuracy alone [1]	ML models outperformed the ESI reference and achieved greater net benefit across clinically relevant thresholds, supporting threshold-based triage decisions [1]
Not provided in retrieved evidence	2022	*JAMA Network Open*	10.1001/jamanetworkopen.2021.10703	Prediction of antithrombotic-related gastrointestinal bleeding	No single threshold rule was proposed; the study reports class-specific operating performance for validation rather than relying on accuracy [2]	AUC, sensitivity, specificity, positive predictive value, and prediction density plots [2]	Machine-learning models offered similar performance to HAS-BLED; class-aware metrics were needed to characterize high-risk prediction [2]
Not provided in retrieved evidence	2022	*PLOS ONE*	10.1371/journal.pone.0269135	Breast-cancer prediction with imbalanced data	No explicit threshold-selection rule; the study compares models on binary-classifier metrics under imbalance [3]	Balanced accuracy, sensitivity, specificity, precision, and F1-score; balanced accuracy, sensitivity, and F1 were highlighted as especially important [3]	Accuracy alone is insufficient in imbalanced medical data; minority-class–sensitive metrics should guide model comparison [3]
Not provided in retrieved evidence	2024	*Journal of Affective Disorders*	10.1016/j.jad.2024.05.078	Depression risk prediction in older adults with chronic disease	Thresholding was not the main focus; models were compared after imbalance-aware training [4]	Accuracy, specificity, sensitivity, PPV, NPV, ROC curve, PR curve, and balanced accuracy; PR curve was noted as more informative under imbalance [4]	PR curves and balanced accuracy help distinguish performance differences that ROC curves can miss in imbalanced settings [4]
Not provided in retrieved evidence	2025	*International Journal of Emergency Medicine*	10.1186/s12245-025-00825-3	ED admission prediction in older adults	Risk thresholds of <0.1, 0.1–0.5, and >0.5 were used to stratify low-, moderate-, and high-risk groups, explicitly balancing sensitivity and specificity [5]	AUC and calibration plots, with decision-curve analysis for clinical utility [5]	Threshold bands can support actionable triage groups and improve disposition planning in older adults [5]

Not provided in retrieved evidence	2025	*Journal of Neurointerventional Surgery*	10.1136/jnis-2024-021759	Clinical outcome prediction in cerebrovascular and endovascular neurosurgery	No single threshold rule; the review argues for rare-event–appropriate evaluation rather than accuracy [6]	AUROC, AUPRC, sensitivity, specificity, precision, and F1; AUPRC is suggested as preferable for rare events [6]	AUROC can be misleading for rare outcomes, while AUPRC, precision, and F1 better reflect clinically meaningful minority-class performance [6]
Not provided in retrieved evidence	2025	*Medical Decision Making*	10.1177/0272989X251343082	Clinical deployment of ML models	Threshold selection is presented as a core implementation step, but no single selection rule is prescribed [7]	Calibration, decision-threshold selection, explainability, bias mitigation, and robust validation; the focus is broader than accuracy [7]	Clinically usable models require threshold choice plus calibration and validation, not just high discrimination [7]
Not provided in retrieved evidence	2025	*Diagnostics*	10.3390/diagnostics15040501	Surgical site infection prediction with highly imbalanced data	Thresholding is not the main reported criterion; the study emphasizes imbalance correction and robust validation [8]	MCC is highlighted as the most reliable summary metric; the study also evaluates model performance under resampling [8]	MCC is presented as more robust than conventional metrics when classes are highly imbalanced [8]
Not provided in retrieved evidence	2025	*Scientific Reports*	10.1038/s41598-025-85121-z	Early sepsis screening in emergency triage	Thresholding is not central in the abstract; the model is framed as a screening tool for early case finding [9]	AUC, F1 score, and calibration curves [9]	Screening performance improved when triage data were expanded beyond vital signs, with F1 and AUC used to judge usefulness [9]
Not provided in retrieved evidence	2026	*Proceedings of SPIE*	10.1117/12.3101077	Single-disease screening for chronic disease risk	Decision thresholds were tuned on precision-recall characteristics rather than fixed at 0.5; validation F1 weighted the ensemble [10]	F1, ROC-AUC, G-Mean, and MCC; threshold optimization is tied to clinical priorities [10]	Threshold optimization on PR behavior produced clinically meaningful gains in recall and F1 across screening tasks [10]
Not provided in retrieved evidence	2026	*JMIR Formative Research*	10.2196/86379	Real-world clinical prediction under class imbalance	No fixed cutoff; the study shows that threshold-insensitive AUROC can hide clinically important changes in error balance [11]	Macro-F1, class-specific precision and recall, calibration, decision curve analysis, and learning curves [11]	AUROC alone is insufficient in imbalanced health care data; routine class-aware metrics are needed [11]
Not provided in retrieved evidence	2026	*Applied Sciences (Switzerland)*	10.3390/app16147263	Emergency triage escalation detection	Validation-selected operating thresholds were chosen to target at least 90% sensitivity [12]	AUROC, average precision, sensitivity, specificity, PPV, calibration, and decision-curve analysis [12]	A sensitivity-targeted threshold can support safe escalation decisions, but recalibration is still needed before direct risk interpretation [12]

Bottom line

Across health screening and triage studies from 2019 to 2026, the dominant pattern is to **avoid accuracy as the main decision metric** and instead use **threshold-aware, minority-class–aware criteria**: F1 or macro-F1 when balancing errors, sensitivity-targeted thresholds for screening, precision-recall–based threshold tuning when false negatives matter, and decision-curve or utility-based selection when clinical consequences are explicit [1] [10] [11] [12]. The most consistently recommended evaluation set is **sensitivity/recall, precision/PPV, F1, PR-AUC/AUPRC, and sometimes MCC or balanced accuracy**, with ROC-AUC treated as helpful but often insufficient under imbalance [3] [4] [6] [8].

Confidence level: **High** — the retrieved health-focused sources repeatedly converge on the same methodological message, with multiple independent examples across screening, triage, and rare-event prediction showing that threshold choice and class-aware metrics outperform accuracy-based evaluation in imbalanced settings [6] [10] [11] [12].

References

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